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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (EVT) (CBCS) Degree Examinations

December 2024 – January 2025
22EVT133 – DIGITAL CONTROLLER

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	List any 8 peripherals of dsPIC33 family controllers and provide their brief details.	08	L*2	1	2,3,5
b)	Explain return from interrupt latency for one cycle instruction and two cycle instruction.	08	L2	2	2,3,5
c)	List any four applications of dsPIC33CH512MP508 controller.	04	L1	1	2,3,5
2. a)	What is modulo addressing mode? Mention how modulo addressing mode is realized in dsPIC33/PIC24 CPU family.	08	L2	1	2,3,5
b)	Identify the functionality of interrupt control register-2 mentioning the role of all the bits.	12	L3	2	2,3,5
3. a)	Write a note on dsPIC33 CPU priority status.	05	L2	1	2,3,5
b)	Explain interrupt setup procedure in dsPIC33 family controllers.	10	L2	2	2,3,5
c)	Explain data memory map of dsPIC33CH512MP508 with block diagram.	5	L2	2	2,3,5
Unit – II					
4. a)	Describe 12-Bit SAR A/D conversion sequence.	06	L2	3	2,3,5
b)	Examine the sampling and conversion timing in dsPIC33 ADC module.	08	L4	3	2,3,5
c)	Asses the modes of operation supported by Timer1 modules of dsPIC33/PIC24 family devices.	06	L4	4	2,3,5
5. a)	Write the configuration and functionality of digital comparator module associated with ADC module.	07	L2	3	2,3,5
b)	Summarize the PWM generator triggers generated in dsPIC33 controllers.	07	L2	4	2,3,5
c)	List the data buffering registers supported by PWM module and provide the different options available for updating the values of internal active registers.	06	L3	4	2,3,5
6. a)	Write a note on common interrupts triggered in ADC module of dsPIC33 family controllers.	06	L2	3	2,3,5
b)	Explain the process of setting 16-bit gated timer mode of Timer1 module in dsPIC33 family controller.	07	L2	4	2,3,5
c)	Explain six different PWM modes supported by PWM module of dsPIC33 family controllers.	07	L2	4	2,3,5
Unit – III					
7. a)	Draw the block diagram for SPI standard mode of operation.	06	L2	5	2,3,5
b)	Explain the master communicating mode in single master environment of I2C module.	10	L2	5	2,3,5
c)	List the four different CAN FD message frames.	04	L1	5	2,3,5
8. a)	Explain the procedure of setting Master mode of operation in SPI module.	08	L2	5	2,3,5
b)	List the different modes of operation of CAN FD module. Also write a note on mode change.	08	L2	5	2,3,5
c)	Write the master message events of I2C module which triggers interrupts.	04	L2	5	2,3,5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
